PUNK ROCK ENGINEERING

Build Fast. Break Things. Learn Everything.

An FRC Engineering Lecture | Inspired by SuperfastMatt

Two Ways to Build Rockets

NASA

"Don't Fail"

- ✔️ Define every requirement
- ✔️ Check every box
- ✔️ Simulate before building
- ✔️ Multiple review gates
- ✔️ Methodical iteration

VS

SpaceX

"Fail Fast, Fail Often"

- ⚡ Take an educated guess
- ⚡ Build it quickly
- ⚡ Strap on sensors
- ⚡ Light it up and see
- ⚡ Learn at warp speed

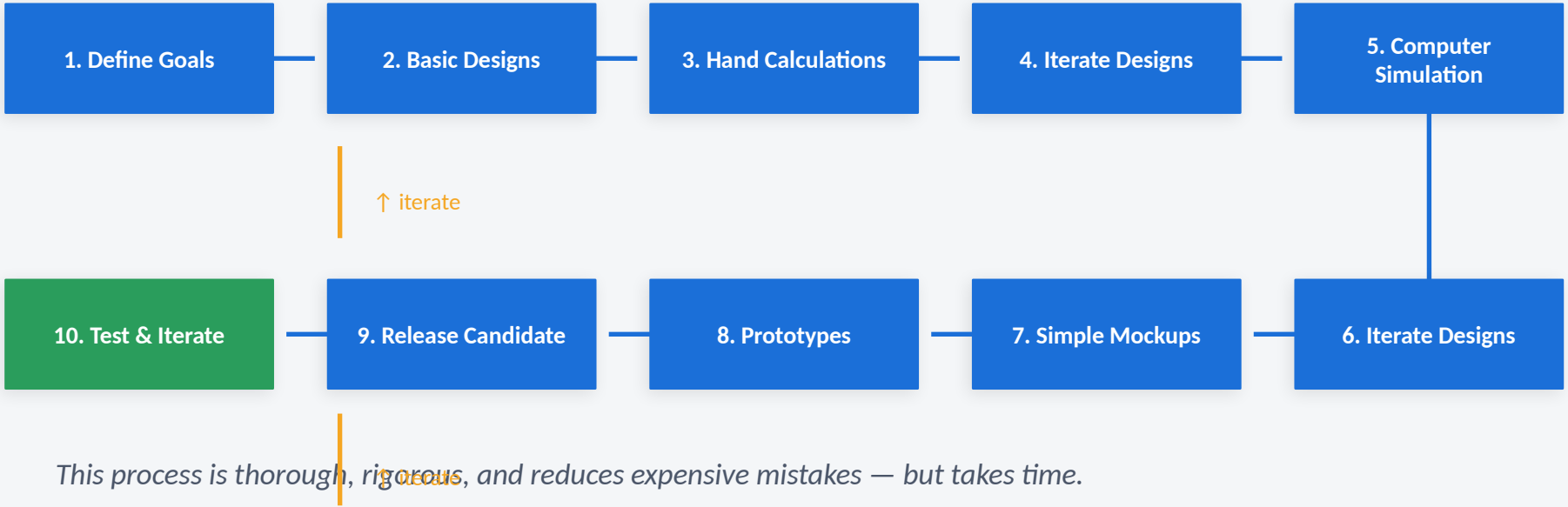
What Is Punk Rock Engineering?

> *"Skip all the boring parts and just sort of guess what the final design will look like, then build it and test it. After which it should be pretty obvious why your design sucks. So you remake it. Then you're done."*

— SuperfastMatt

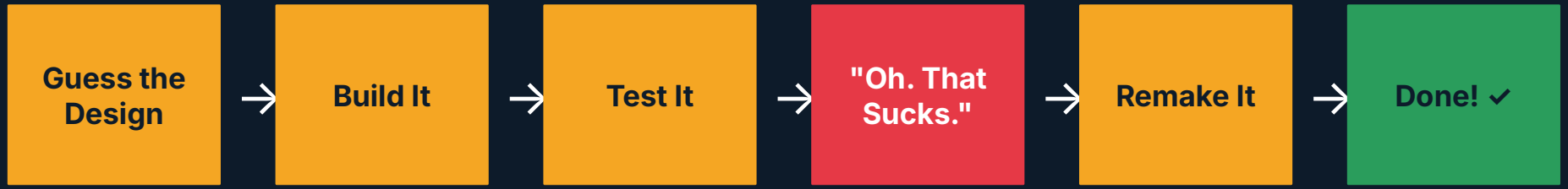
The punk rock ethos applied to engineering: skip the process, trust your gut, build it, find out it sucks, fix it.

The Traditional Engineering Design Cycle



This process is thorough, rigorous, and reduces expensive mistakes — but takes time.

The Punk Rock Engineering Cycle



Engineering intuition

Whatever you can

See what happens

This is the moment of truth

Now you know what you want

Ship it!

The whole cycle can take hours — not months.

Disclaimer: "intuition" here is doing a LOT of heavy lifting.

⚠ The Big Caveat

Engineering intuition is to engineering...

...as astrology is to astronomy.

— SuperfastMatt

Astrology

Looks like science
Uses made-up patterns
No predictive rigor
Feels right, might be wrong

Astronomy

The real deal
Testable hypotheses
Measured, verified
Predictable results

Punk rock engineering isn't fake — but your intuition must be earned through experience.

Side by Side: Two Approaches

Category	Traditional Engineering	Punk Rock Engineering
Starting point	Clear requirements + specs	Gut feeling + rough idea
First action	Research & calculate	Build something
Learning style	Simulate before building	Build to find out
Risk tolerance	Minimize failure	Embrace failure early
Iteration speed	Slow & deliberate	Fast & frequent
Requires	Specs, time, discipline	Tools, materials, intuition
Best for	High stakes, large teams	Getting unstuck, learning

Why Building Early Can Be Smarter

There are things you simply cannot learn until you dive in.

Complex Part Interactions

How your intake, conveyor, and shooter actually work together can't be fully simulated — only discovered.

Unknown Unknowns

You don't know what you don't know. A physical prototype reveals constraints you never would have thought to simulate.

Clarifying Your Goals

"You might not know what you really want until you go for it and find out that's not what you want." — SuperfastMatt

This is why prototyping fast is a legitimate engineering strategy — not just being impatient.

In FRC: When to Use Which Approach

Traditional Engineering

- 🔵 Drivetrain chassis design
- 🔵 Structural load calculations (arm, elevator)
- 🔵 Game strategy and game piece analysis
- 🔵 End-of-season design review for next year
- 🔵 Anything that mounts motors or bears load

Punk Rock Engineering

- ⚡ Intake geometry (how many tries does it take?)
- ⚡ Shooter angle and flywheel speed
- ⚡ Gripper / end effector design
- ⚡ Early-season brainstorming builds
- ⚡ "Does this concept even work?" tests

✅ WHEN TO GO PUNK ROCK



You're stuck

Overthinking it? Just build a rough version. You'll learn more in 20 minutes of testing than 2 hours of agonizing.



Personal / side projects

No budget review. No safety audit. Just you and a problem. This is where punk rock thrives.



Early prototyping

Week 1–2 of build season. The goal is to test if an idea is viable at all, not to build the final robot.



When time is short

Something's broken at 9pm before a Saturday competition. Punk rock engineering is the only option.



Learning something new

Building a swerve drive for the first time? Build a bad one first. There's no shortcut to the first one.

🚫 WHEN NOT TO GO PUNK ROCK



Your final competition robot

You've tested concepts. Now apply rigor. A robot that fails at competition due to sloppy design costs you qualification points.



Your real job (eventually)

Boeing, SpaceX, your local hospital. Society depends on engineers who follow the process. Save punk rock for personal projects.



Load-bearing / structural systems

Elevators, arms, climbers. If it could hit someone when it fails, run the calculations.



Expensive or one-of-a-kind parts

Custom machined parts, sponsored hardware, anything that takes 2 weeks to replace. Simulate first.



Electrical systems & wiring

Shorts, heat, fire. The engineering process exists here for very good reasons.

The Real Takeaway

01

Hesitate less.

Overthinking kills momentum. If you're stuck, build something — even if it's wrong. The build will tell you what to do next.

02

Prototype more.

Every prototype teaches you something a simulation won't. The physical world is the ultimate test environment.

03

Know when to switch gears.

Punk rock is for exploration. Traditional engineering is for execution. Great engineers do both — and know which mode they're in.

Both approaches are engineering. Neither one alone makes a great engineer.

Discussion & Activity

Group Exercise — 5 Minutes

1. Think of a time you were stuck on a design decision. What was holding you back? How could a quick prototype have helped?

2. This year's game: pick one subsystem. Which approach would you use first — traditional or punk rock? Why?

3. SuperfastMatt says he'd "rather just build something and see what happens, even knowing he might have to build it twice." Agree or disagree?

Then: as a team, write one sentence: "We will use punk rock engineering when _____."